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Competition Details

Summary

ITEM	SPECIFICATION
Enclosed flight area	6.0 m length × 4.0 m width × 3.0 m height
Drone platform	Ryze Tello / DJI Tello EDU
Primary vision sensor	Forward facing camera
Fire targets	3 red circular markers
Decoys / Distractors	4 non-target markers with different colours and shapes
Marker mounting	Printed markers mounted vertically on stable supports or tripods
Marker centre height	2.0 m above the floor
Take off / landing	Fixed floor position at coordinate (3.0, 3.0)
Programming	Block coding interface or Python
Manual control during scored run	Not permitted

Coordinate System

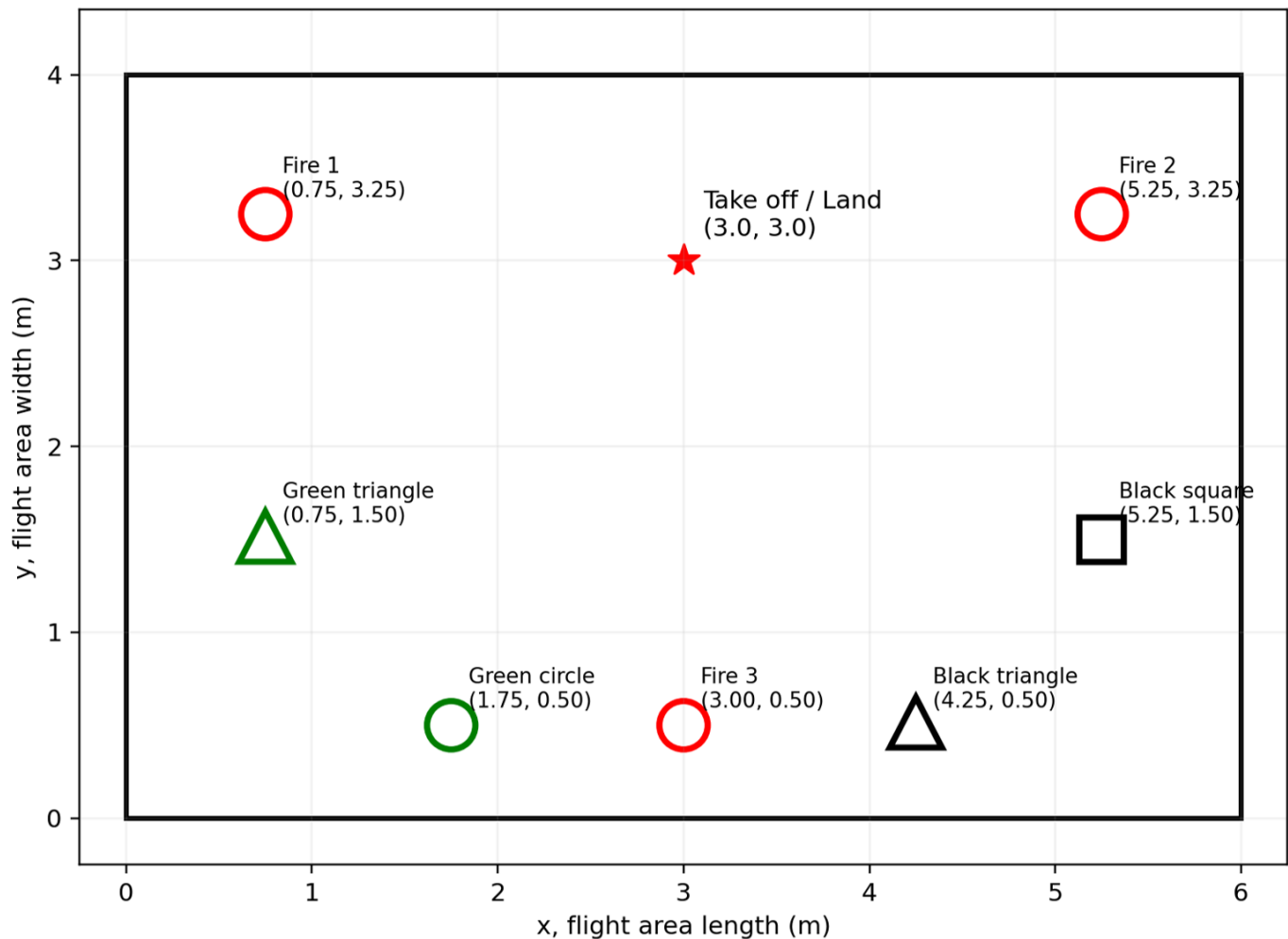
The flight area is treated as a local Cartesian coordinate system. The x axis follows the 6 m length, and the y axis follows the 4 m width. The takeoff and landing position is located at (3.0, 3.0). Marker coordinates refer to the center position of each printed target when viewed from above. All marker centers are initially proposed at a height of 2.0 m so that the visual classification task is separated from unnecessary altitude complexity.

Target and Decoys Positions

MARKER	VISUAL CLASS	X (M)	Y (M)	Z (M) / CENTER HEIGHT	REQUIRED RESPONSE
Fire 1	Red circle	0.75	3.25	2.0	Detect, approach and flip
Fire 2	Red circle	5.25	3.25	2.0	Detect, approach and flip
Fire 3	Red circle	3.00	0.50	2.0	Detect, approach and flip
Decoy 1	Green triangle	0.75	1.50	2.0	Ignore
Decoy 2	Black square	5.25	1.50	2.0	Ignore
Decoy 3	Green circle	1.75	0.50	2.0	Ignore
Decoy 4	Black triangle	4.25	0.50	2.0	Ignore

Environment Layout

The layout deliberately includes targets with different colours and shapes. The green circle markers are intended to test the colour recognition, while other noncircular targets are intended to test shape recognition. The physical positions can be reproduced directly from the previous table. All seven marker centres are fixed at 2.0 m above the floor.



All marker centres are positioned at 2.0 m above the floor.

Competition Rules

The objective is to find all three simulated fires represented with red circle markers while avoiding all decoy markers. Teams may choose their own search strategy and flying path. The provided marker positions provide a consistent physical setup, but the challenge remains the correct visual identification and autonomous response.

- The drone take-off and landing position are at (3.0, 3.0).
- The drone must take off autonomously and begin its developed search strategy.
- When a marker is seen by the forward-facing camera, it must be determined whether it is a red circle or not.
- If the marker is a red circle, the drone must approach it to the required detection position and perform one flip to signal that a fire has been found.

- If the marker is not a red circle, the drone must ignore it and continue the search without performing a flip.
- Each of the three fire targets should be counted once only. Repeated flips at the same fire do not create additional detections.
- A flip performed in response to a decoy is recorded as a false positive.
- After all, three fires have been correctly identified, the drone must return to the take-off and landing position and land.
- Once the scored run begins, manual piloting or manual correction is not permitted.

Programming and Computer Vision

You will be provided with the competition block coding interface. The interface will allow you to build the mission using visual programming blocks while Python functions operate in the background. This keeps the activity accessible to everyone who have limited text based programming experience while still exposing them to real computer vision and autonomous decision making.

- Teams may use the block coding interface or choose to program directly in Python.
- Both pathways must complete the same physical mission and are assessed using the same competition rules.
- The vision system must distinguish the red circular fire targets from the four decoys.
- Teams may use colour detection, shape detection, contour analysis, image classification, or another suitable approach available within the competition environment.
- The search route itself may use programmed movements and rotations. Computer vision is used to decide whether the object being observed is a valid fire target.

Assessment Framework

Each team may complete up to three official attempts, and the best result achieved across those attempts will be used for assessment. The winner is determined by target classification accuracy. Each correctly identified fire target is worth 2 points and each misidentification is penalised by 2 points. If two or more teams achieve the same best score, the team with the shortest total flight time from take-off to landing on their best scoring attempt ranks higher.

SCORING EVENT	POINTS	RULE
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Correct fire identification	+2	Awarded when the drone correctly identifies one of the three red circles as a fire.
Misidentification	-2	Applied when the drone identifies a distractor as a fire.
Tie breaker	Flight time	If teams have equal best scores, the shortest elapsed time from take off to landing on the best scoring attempt wins.

The maximum classification score is 6 points if all three fires are correctly identified and no distractors are misidentified. Each team may make up to three official attempts, with its best scoring attempt retained. Flight time is used as the tie breaker when teams finish with the same best score.

Preparation and Testing

- The block coding interface and technical instructions will be provided before the competition.
- Teams choosing Python may develop and test their code directly.
- The marker specifications, coordinate layout and example images should be supplied before the event so teams can prepare.
- Teams with access to their own compatible Tello drone may begin testing after the competition materials are released.
- Competition organisers may provide Tello drones for supervised testing before the competition, with final arrangements communicated closer to the event.

Safety and Operational Rules

All competition flights will occur inside the enclosed flight area. A group of specialised professionals will manage operational procedures, preflight checks, battery management, supervision and emergency procedures for the event.

The competition specification defines the participant challenge and does not replace the specialised professionals operational procedures or the event specific risk assessment.

Prizes

TBA

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